

Yu (Adam) Ding

Postdoctoral Research Fellow ■ Machine Learning for Healthcare ■ Cancer Data Science

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Research Summary

Ph.D.-trained researcher developing machine learning and optimization methods for healthcare data analysis. Current work focuses on intratumor heterogeneity, cancer evolution, and multi-omics integration, with emphasis on statistically grounded and computationally efficient models for translational oncology.

Professional Experience

The University of Texas MD Anderson Cancer Center

Houston, Texas

Postdoctoral Research Fellow

2024 – Present

The University of Kansas School of Business

Lawrence, Kansas

Research Assistant in Operations Management

2019 – 2020

Education

Binghamton University

Vestal, New York

Ph.D. in Industrial and Systems Engineering (Machine Learning for Healthcare)

2020 – 2024

Wayne State University

Detroit, Michigan

M.S. in Industrial Engineering (Reliability / Machine Learning)

2017 – 2019

University of Science and Technology of China

Anhui, China

B.S. in Geophysics (Quantitative Methods in Solid Geophysics)

2013 – 2017

Publications

J: Journal C: Conference D: Dissertation *: Corresponding author

- [J.7] Ding, Y., Costa, C., Si, B.* (2026). **Federated function-on-function regression with an efficient gradient boosting algorithm for privacy-preserving telemedicine**. *IEEE Transactions on Automation Science and Engineering*, 23, 4704–4716. **QCRE Best Student Paper Finalist, IISE 2024**. [video]
- [C.1] Jahedi, A.*, Unal, M., Ding, Y., Nurmammadova, A., Liu, Y. (2025). **Boundary-Specific Remodeling Defines Divergent TLS Maturation in Colorectal Cancer**. *Allison Institute Scientific Symposium*.
- [J.6] Ding, Y., Somers, V., Si, B.* (2025). **A novel sparse generalized structural equation modeling with structured sparsity for subgroup discovery from multi-modal mixed-type data**. *IISE Transactions*, 57(12), 1497–1511.
- [J.5] Sutherland, M. A.*, Hutchinson, M. K., Si, B., Ding, Y., Liebermann, E., Connolly, S. L., ... Mueller, S. D. (2025). **Health screenings in college health centers: Variations in practice**. *Journal of American College Health*, 73(8), 3031–3038.
- [D.1] Ding, Y. (2024). **Statistical Machine Learning and Data Fusion Methodologies: Applications in Healthcare**. Binghamton University.
- [J.4] Mueller, S. D.*, Sutherland, M. A., Hutchinson, M. K., Si, B., Ding, Y., Connolly, S. L. (2024). **Student Health Services at Historically Black Colleges and Universities and Predominantly Black Institutions in the United States**. *Health Equity*, 8(1), 226–234.
- [J.3] Alramadeen, W., Ding, Y., Costa, C., Si, B.* (2023). **A novel sparse linear mixed model for multi-source mixed-frequency data fusion in telemedicine**. *IISE Transactions on Healthcare Systems Engineering*, 13(3), 215–225. **Feature Article, ISE Magazine**.
- [J.2] Jiang, L., Ding, Y., Sutherland, M. A., Hutchinson, M. K., Zhang, C., Si, B.* (2022). **A novel sparse model-based algorithm to cluster categorical data for improved health screening and public health promotion**. *IISE Transactions on Healthcare Systems Engineering*, 12(2), 137–149.

[J.1] Ding, Y., Yang, Q.*, King, C. B., Hong, Y. (2019). **A general accelerated destructive degradation testing model for reliability analysis**. *IEEE Transactions on Reliability*, 68(4), 1272–1282.

Manuscripts

M: Manuscript *: Corresponding author

- [M.2] Mueller, S.*, Ding, Y., Si, B., Sutherland, M., Hutchinson, K. (2026). **Access to Campus Health Services at MSI and Non-MSI Colleges and Universities in the U.S.**. Minor revision *Journal of Health Equity*.
- [M.1] Ding, Y., Costa, C., Si, B.* (2025). **Vertical Federated Learning of Gradient Boosting for Functional Regression with Differential Privacy**. Submitted to *IEEE Transactions on Privacy*.

Honors and Awards

Warren Alpert Computational Biology and AI Network Fellow	Aug 2025
<i>UCLA Computational Genomics Summer Institute</i>	
Best Team Award	Apr 2025
<i>James P. Allison Institute and IDSO Hackathon</i>	
QCRE Best Student Paper Competition Finalist	May 2024
<i>IISE Annual Conference & Expo</i>	
National Science Foundation (NSF) Travel Award	May 2024
<i>The University of Arizona</i>	
Watson Professional Development Fund	Apr 2024
<i>SUNY Binghamton</i>	
Summer Research Fellowship	Apr 2020
<i>The University of Kansas</i>	
University Graduate Fellowship	Aug 2019
<i>The University of Kansas</i>	
Graduate Fellowship	Aug 2018
<i>Wayne State University</i>	
National Encouragement Scholarship	Aug 2016
<i>University of Science and Technology of China</i>	
Kansas Half Marathon Finisher Prize	Oct 2019
<i>Kansas Half Marathon</i>	

Conference Presentations and Invited Talks

- **Federated function-on-function regression with an efficient gradient boosting algorithm for privacy-preserving telemedicine**, Department of Radiation Oncology, Mayo Clinic Arizona, Phoenix, AZ, July 1, 2024.
- **Federated function-on-function regression with an efficient gradient boosting algorithm for privacy-preserving telemedicine**, IISE Conference, Montreal, Canada, May 18, 2024. [\[video\]](#)

Teaching Experience

Rice University	Spring 2025
<i>Guest Lecturer</i>	Lecture recording
▪ Statistics 623/423: Probability in Bioinformatics and Genetics (lectures, review sessions, exam questions, office hours)	
Binghamton University	Spring 2024
<i>Guest Lecturer</i>	
▪ SSIE 548 / ISE 448: Healthcare Data Science & Analytics (12 lectures, reviews, Q&A)	
▪ Teaching evaluation score: 4.5 / 5.0	

Teaching Assistant

- IE 6430: Computer Simulation Methods (graduate)

Wayne State University

Fall 2017

Teaching Assistant

- IE 7270: Reliability Estimation (graduate)

Grant Writing Experience

Grant preparation with Ph.D. advisor Dr. Bing Si

Binghamton University

- NIH/NHLBI R01HL168173: *Sleep and Cardiometabolic Subgroup Discovery and Risk Prediction in U.S. Adolescents and Young Adults*. Amount: \$2,452,065.
- NIH/NHLBI R21HL161765: *Towards Precise Phenotype Discovery of Obstructive Sleep Apnea with a Data-Inclusive Multi-Study Analysis*. Amount: \$242,770.
- SUNY-IBM AI Research Alliance: *Ray-F2R-FL: Ray-based Functional Regression with Federated Learning*. Amount: \$200,000.

Student-Led Research Grant Proposal

The University of Kansas

- Independently wrote a student-led research grant proposal. Amount: \$5,000.

Professional Service

- Journal reviewer — *Nature Reviews Cancer*
- Journal reviewer — *Journal of the American Statistical Association*
- Journal reviewer — *IEEE Transactions on Medical Imaging*

Professional Memberships

- Institute of Industrial and Systems Engineers (IISE)
- INFORMS (Institute for Operations Research and the Management Sciences)
- IEEE (Institute of Electrical and Electronics Engineers)